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Job Search, Hours Restrictions, and Desired Hours of Work

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A structural empirical job search model is presented that incorporates the labor supply decision of individuals. The arrival of a job offer is modeled as a random draw from a wage-hours offer distribution. Subjective information is used on desired working hours to identify optimal hours from offered hours. Policy simulations are performed to address several policy questions: Does a decrease in unemployment benefits lead to the acceptance of jobs with less preferred working hours? How does a decrease in the length of the standard working week affect the job acceptance behavior of the unemployed?

I. Introduction

In this article an empirical structural job search model is specified in which I explicitly incorporate working hours as a job characteristic that influences the job acceptance decision of individuals. Thus this approach brings together elements from the labor supply literature and the job search literature.

Early empirical studies in the field of labor supply (e.g., Heckman 1974; and Hausman 1980) typically focused on the neoclassical labor supply model. In these studies the labor supply function, defined by the tangency point of the indifference curve and the budget constraint, is parameterized. Data on observed weekly hours of work are used to estimate the parameters of the labor supply function.¹

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¹ See Blundell and MaCurdy (1999) for an overview of labor supply models.

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The neoclassical model typically ignores restrictions on working hours that may stem from the demand side of the labor market: in the empirical implementation of the model it is assumed that there are no systematic differences between observed working hours and optimal working hours. The model does not account for “involuntary” unemployment.² Moreover, the neoclassical model does not incorporate that the number of jobs available is limited, that hours in a job offer are often fixed by the employer, and that it takes time to search for and find an acceptable job. Altonji and Paxson (1992) show that working hours are affected more strongly when the job changes than when the job stays the same, which is consistent with the existence of hours restrictions.³ In the empirical literature it was noticed that the neoclassical labor supply function cannot reasonably mimic the peak in the frequency of observed working hours at institutionally determined levels of working hours like 40 hours a week.

This created an incentive to collect subjective survey information on “desired” weekly working hours. Ham (1982) and Blundell, Ham, and Meghir (1987) use survey information on underemployment as additional information in estimating labor supply. Stewart and Swaffield (1997) use information on both underemployment and overemployment and allow for differences in the process of observed working hours depending on the regime. Kahn and Lang (1991) use information on desired weekly working hours to test for systematic differences between observed working hours and desired working hours. Other studies took the neoclassical labor supply model as the point of departure but estimated it with subjective data on preferred hours instead of observed hours.⁴

An alternative development in the literature is described in a series of studies (Van Soest, Woittiez, and Kapteyn 1990; Tummers and Woittiez 1991; Dickens and Lundberg 1993; Bloemen 2000). Job offers and hours restrictions are explicitly incorporated in the model. This is done by assuming that individuals receive a random number (possibly zero) of job offers. A job offer is characterized by a wage rate and a weekly number

² Ham and Reilly (2002) test the labor supply model in which hours only depend on wages and taste shifters, which holds if any shocks from the demand side of the labor market are incorporated into wages. They conclude that demand side factors enter the labor supply equation, in addition to the wage. They provide alternative model specifications that motivate the appearance of demand side factors in the hours equation, like an implicit contracts model and an hours restriction model. The latter model allows for involuntary unemployment by assuming that desired working hours may differ from observed working hours in case of unemployment.

³ Euwals (2001) did a comparable study for the Netherlands.

⁴ An example of this approach is Woittiez and Kapteyn (1998). The focus in Woittiez and Kapteyn (1998) is on the incorporation of interdependent preferences and habit formation, as an alternative extension of the standard neoclassical labor supply model.

of working hours. Individuals select the job with the highest utility level, whereas staying jobless is also an option.⁵ Cross section data on observed working hours and wages are used to estimate the model. The hours offer distribution is modeled as a discrete distribution. As a consequence the models do rather well in mimicking the empirical distribution of observed weekly working hours.

Identification is a serious problem in static labor supply models with hours restrictions and job offers. Data on hours and wages are used to identify the parameters of preferences for work, the number of job offers, the hours offer distribution, and the wage distribution. Therefore there are two major identification problems. First, the estimation of the job offer probability proceeds without the use of any information on numbers of job offers (or, alternatively, on time elapsed before a job offer or a transition occurs).⁶ Second, both offered hours and optimal hours are identified from data on observed working hours only.

To address the identification of the job offer arrival probability, note that applications of job search theory use data on unemployment duration to empirically identify job offer arrival rates.⁷ However, in the job search framework the wage usually is the only job characteristic. There are still very few job search models in the literature that incorporate working hours, and the models that do account for working hours usually assume that working hours can be chosen unrestrictedly by the individual. Burdett and Mortensen (1978) introduced a model of job search and optimal working hours and search time. Bloemen (1997) estimates a structural job search model in which hours can be chosen optimally. In this article we expand on the empirical literature of structural job search models by formulating a job search model with hours restrictions. We show that the resulting model has a reservation utility property: all combinations of wage rates and working hours with a utility level higher than the reser-

⁵ Yet a different class of labor supply models, that is related to models with hours restrictions, is used by MaCurdy, Green, and Paarsch (1990) and Hoynes (1996). In this approach the optimal hours is the outcome of a discrete choice process (e.g., between part-time and full-time work). The observed working hours are modeled as a continuous variable: they are assumed to be equal to the (discrete) optimal working hours up to a multiplicative (continuously distributed, for instance, lognormally) error term.

⁶ Bloemen (2000) illustrates the identification problem by introducing observed heterogeneity in the job offer arrival rate. It results in a distribution of optimal working hours that is approximately uniform over the relevant range of hours. This means that no information on preferences can be extracted from the data anymore once it is allowed for more flexibility in the job offer probability.

⁷ A job offer arrival rate that depends on observed individual characteristics is quite common in empirical implementations of job search models since the work by Narendranathan and Nickell (1985). See Mortensen (1986) for an overview of the basic elements of job search theory.

vation utility level are acceptable to the individual. Alternatively, we can show that the reservation wage rate is a function of working hours that takes a minimum if hours are optimal. For each hours level we can evaluate the reservation wage rate and the resulting job acceptance probability.

To address the identification of the job offer distribution subjective information on desired weekly working hours is used as a source of information in addition to observed working hours. I show what is the differential impact of adding this information on desired working hours to illustrate its impact for the identification of the underlying behavior. As mentioned above, subjective information on working hours has been used before in the estimation of structural models for the reason that observed working hours are influenced by the demand side of the labor market, but here this information is used to actually separate preferences for hours from hours offers. In the analysis I use data from the Dutch Socio-Economic Panel, a household survey in the Netherlands.

Understanding the role of hours restrictions is potentially very important to assess the effects of some labor market reforms that have been at the center of the political debate in recent years. A model that contains the unemployment hazard, hours restrictions, and preferences is a very useful tool to address several policy issues concerning the relation between hours restrictions, job acceptance behavior, and the distribution of accepted working hours. For example, does a decrease in unemployment benefits lead to the acceptance of jobs with unfavorable working hours according to the individual preferences? To address this question we simulate the impact of a decrease in benefits by 5% on the distribution of accepted working hours. How does a decrease in the length of the standard working week affect the job acceptance behavior of the unemployed? Does the expected unemployment duration decrease if the hours offer distribution is more similar to the distribution of desired hours, or will that lead to an increase in the value of search such that unemployment duration increases? To address these questions we will simulate the impact of a change in the hours offer distribution on the unemployment hazard and the distribution of accepted working hours.

In the next section the model is presented. Section III provides a description of the available data. Section IV contains the estimation results and the outcomes of the policy simulations. In Section V, I present conclusions.

II. The Model

In Section II.A, I formulate the job search model with hours restrictions. In Section II.B, I choose functional forms and distributions of error terms that are used in the empirical specification. In Section II.C, I show how the subjective data on desired working hours are incorporated in the

model. In Section II.D, I discuss the formation of the likelihood contributions that are necessary for the estimation of the model parameters by maximum likelihood.

A. Optimal Strategy

Individuals maximize the expected present value of utility, subject to a budget constraint, while incorporating expectations about future job offers and layoffs. Utility in period t is specified by a utility function that is defined over net weekly income y_t and weekly working hours h_t .⁸ The utility function may include random preferences, which are denoted by ϵ .⁹ Period t utility is denoted by

$$u(y_t, h_t; \epsilon) \text{ with } \frac{\partial u(y_t, h_t; \epsilon)}{\partial y_t} > 0. \quad (1)$$

The budget constraint is linear. For someone employed it defines weekly income as the sum of labor income and nonlabor income μ : $y_t = w_t h_t + \mu_t$.¹⁰ For someone unemployed income consists of weekly benefits b_t and (weekly) nonlabor income: $y_t = b_t + \mu_t$.

Unemployed individuals receive job offers with constant arrival rate λ . This arrival rate may depend on unobserved random heterogeneity, denoted by v . Accordingly, we denote $\lambda = \lambda(v)$. The unobserved heterogeneity v represents unobserved or unobservable individual characteristics, that affect the probability of receiving a job offer. In duration models for unemployment, it has become common practice to include unobserved heterogeneity in the hazard rate (see Lancaster 1979) to capture the possibility that individuals with the same observed components can have different escape rates out of unemployment due to unobserved characteristics (or abilities).¹¹ Thus we allow for the fact that, after allowing for

⁸ Note that utility is usually defined over consumption instead of income. Here I have replaced consumption by the available budget. I do not model savings behavior.

⁹ Randomness in preferences is assumed to arise from individual characteristics that are observed by the agent but unobserved by the econometrician.

¹⁰ Note that the assumption of a linear budget constraint is a simplification. In particular the work by Hausman (1980) focuses on the incorporation of nonlinear budget constraints due to the taxation of income. Also in a model with hours restrictions nonlinearities in budget constraints can be dealt with; see, e.g., Bloemen (2000). However, in the present study we want to focus on the identification problem of preferences and offered hours, which is the additional complication that arises in this class of models as compared to empirical models that are solely based on the neoclassical model. Therefore we abstain from nonlinearities in the budget constraint.

¹¹ One can think of various individual characteristics that may influence the arrival rate, ranging from unobserved human capital specificities to social skills, e.g., the individual style of presenting oneself at a job interview.

observable characteristics, the unemployment durations of individuals may be different due to unobservable characteristics.

Furthermore, let q be a two-dimensional vector that contains both sources of random variation, that is, random preferences ϵ and unobserved heterogeneity in the arrival rate v : $q = (\epsilon, v)'$. For employed individuals there is an exogenous constant layoff rate δ .¹²

A job offer, as it is offered by an employer to an unemployed job searcher, consists of an hourly wage rate w_i and a weekly amount of working hours h_i . A job offer is assumed to arrive randomly from a joint wage-hours offer distribution, denoted by $f(w, h)$:

$$f(w, h) = f(w)p_l, 0 < w < \infty, h = h_l, l = 1, \dots, L. \quad (2)$$

In (2) hours offers are modeled with a discrete distribution function: $p(h = h_l) = p_l$. In the empirical specification, the hours offer probabilities are the same for all individuals, and this assumption may be restrictive. However, it is imaginable that the distribution of offered hours depends on institutional values for working hours. Thus, we assume that optimal working hours are determined by preferences and hence depend on observed and unobserved taste shifters and follow a continuous distribution, whereas the distribution of offered working hours is discrete, representing the restrictions faced by the individual.

In appendix A we show that the individual's optimal acceptance strategy is characterized by a reservation utility level $\bar{u}(q)$:

$$\begin{aligned} \bar{u}(q) = & u(b + \mu, 0; \epsilon) \\ & + \frac{\lambda(v)}{\rho + \delta} \sum_{l=1}^L p_l \int_{\xi(h_l, \bar{u}(q); \epsilon)}^{\infty} [u(wh_l + \mu, h_l; \epsilon) - \bar{u}(q)] f(w) dw. \end{aligned} \quad (3)$$

In (3), $\xi(h_l, \bar{u}(q); \epsilon)$ represents the hours dependent reservation wage: jobs with hours h_l and a wage rate exceeding $\xi(h_l, \bar{u}(q); \epsilon)$ are acceptable to the individual. For a given utility level $\bar{u}(q)$ the reservation wage rate is lower the closer is the hours level h_l to optimal hours. Here optimal hours is the level of hours for which the marginal rate of substitution between hours and income is equal to the reservation wage rate.¹³ Thus, individuals

¹² In the empirical specification, the layoff rate depends on observed characteristics. However, we do not include unobserved heterogeneity.

¹³ Note that working hours affect utility both indirectly, by the budget constraint, and directly. Thus, working hours that are higher than optimal hours involve a positive effect on utility through income and a negative effect on utility by a smaller amount of leisure, whereas for hours levels below the optimal level, there is a negative effect on utility by a lower income level and a positive effect by a larger amount of leisure. Thus, by this nature, economic theory imposes no a priori symmetry between positive and negative deviations from the optimal hours.

will only accept jobs with unfavorable working hours if they are compensated sufficiently in terms of the wage. The subjective rate of discount is indicated by ρ . In the empirical application we will fix ρ such that the yearly discount rate is equal to 5%, which is close to the real rate of interest in the Netherlands.

The second term at the right-hand side of (3) represents the expected gains from search. Note that individuals whose optimal hours have a high offer probability (for a given value of the job offer probability) will have higher gains of search and consequently will more often reject an offer and continue searching than individuals whose optimal hours have a low probability of being offered.

The resulting expression for the escape rate out of unemployment is

$$\theta(q) = \lambda(v) \sum_{i=1}^L p_i \bar{F}(\xi_i(q)), \quad (4)$$

in which $\bar{F}(\cdot) = 1 - F(\cdot)$ is one minus the distribution function of offered wages, and $\xi_i(q) \equiv \xi(h_i, \bar{u}(q); \epsilon)$.

B. Specification

We specify the utility function according to Hausman (1980):

$$u(y, h; \epsilon) = -\ln(\gamma - \beta h) - \frac{\beta(h - X\alpha - \epsilon - \beta y)}{\gamma - \beta h}, \quad \beta < 0, \gamma > 0, \quad (5)$$

where X is a vector of observed individual characteristics that may influence preferences.¹⁴ Utility function (5) combined with a linear budget constraint implies a labor supply function that is linear in the wage rate and in nonlabor income:¹⁵

$$h^*(w, \mu) = \mu\beta + w\gamma + X\alpha + \epsilon. \quad (6)$$

The choice of utility function (5) is made partly because it has been applied widely within the literature of labor supply but also for reasons of trac-

¹⁴ Note that the inclusion of observed characteristics in the utility framework is another extension on the common practice in the empirical structural job search model, in which the utility function is usually equal to the (log)wage. For instance, marital status is often included as a taste shifter in labor supply models, but in empirical job search models it is not a variable that is usually assumed to affect the wage offer distribution and the job offer arrival rate. A utility function like (5) motivates the inclusion of a variable like marital status.

¹⁵ It follows from the model that the labor supply function $h^*(w, \mu)$ is positive if the wage w is larger than the reservation wage $\xi(h_b, \bar{u}(q))$ for any hours level h_b . This can be seen by noting that the reservation wage rate at optimal hours h^* is always larger than the virtual wage rate $w^*(q)$, that is implicitly defined by $h^*(w^*(q), \mu) = 0$, while the reservation wage rates at other hours levels are higher than at the optimal hours level.

tability. For example, in combination with the specification of the wage offer distribution below, the gains of search in (3) can be solved analytically, which is desirable for reasons of computational feasibility. Moreover, the existence of an explicit solution for the labor supply function (6) makes it possible to easily combine subjective data on desired hours with the optimal hours generated by the model. Even small generalizations of the labor supply function may complicate the analysis considerably since for solving the reservation utility level we need the direct utility function. Bloemen and Kapteyn (forthcoming), in the context of a static neoclassical labor supply model with taxes, use a labor supply function that is quadratic in the wage rate and show how it complicates the analysis because of the need to impose coherency restrictions on preference parameters and the wage distribution.

Wage rates arrive from a lognormal wage offer distribution with log variance τ and log mean $\zeta'x$.¹⁶ We allow for a lognormally distributed measurement error in the observed wage. Thus, we allow for a difference between the latent accepted wage offer w and the observed wage w_{obs} .

The hours are categorized in 21 classes of 4 hours, with hours ranging from 1 to 84. The job offer arrival rate is parameterized as $\lambda(v) = \exp(\kappa_u'z_u + v)$ and the layoff rate as $\delta = \exp(\kappa_e'z_e)$, where z_u and z_e are vectors of observed characteristics and κ_u and κ_e are parameter vectors.

Unobserved heterogeneity q follows a bivariate normal distribution. Its density function is denoted by $g(q, \Sigma)$; Σ represents the covariance matrix and contains the variance σ_ϵ^2 of random preferences ϵ , the variance σ_λ^2 of the random term v in the job offer arrival rate, and the covariance between the two $\sigma_{\epsilon v}$.

C. Specification of Desired Working Hours

For specifying the relation between subjective information on desired hours h and optimal hours generated by the model we will employ various specifications. I consider three versions of the model.

I call the first model the “base model.” It assumes that observed hours follow from the structural search model that was outlined in Section II, but in the estimation I do not use the information on desired hours. The base model serves as a reference case for the other two model variants, and a comparison of model outcomes will show us the differential impact of the use of information on desired working hours, in addition to information on observed working hours.

The second model variant will be referred to as the “optimal desired hours model.” In this model variant I incorporate information on desired hours in the estimation of the model. The structure of the economic model

¹⁶ Note that the model can easily be extended by making the mean of the wage offer distribution dependent on the number of working hours.

though remains the same: observed hours are modeled as the realization of an accepted wage-hours offer. I denote the reported desired number of working hours by $\tilde{h}$. Desired working hours are linked to optimal working hours h^* , defined in (6), by

$$\ln \tilde{h} = \ln h^* + \nu, \nu \sim N(0, \sigma_\nu^2). \quad (7)$$

In (7) ν represents a normally distributed measurement error. Thus, (7) establishes a close link between the observed desired hours and the neo-classical optimal working hours. Consequently, the density $r(\tilde{h}|w, q)$ of observed desired hours conditional on the wage and unobserved heterogeneity is the lognormal density implied by (7).

The third model I call the “hours satisfaction model.” In this final model variant I incorporate information on desired hours in a more fundamental way. Here I explicitly incorporate the phrasing of the questionnaire, described in Section III. In the questionnaire the respondents are first asked whether or not they are satisfied with current working hours. If they are not satisfied, they are asked to report their desired number of working hours $\tilde{h}$. If they are satisfied, the questionnaire implicitly assumes that their desired number of working hours is equal to the observed working hours. In the model, I assume that the respondents will answer that they are not satisfied with current working hours if the difference between observed hours and optimal hours is “large.” If the difference between observed working hours and optimal working hours is not “too large,” they will answer that they are satisfied with the current working hours. The idea behind this specification is that “satisfaction” with current working hours is not necessarily the same as “optimality”: thus, I do not assume that observed hours are equal (up to an error term) to optimal hours if respondents report being satisfied with working hours, as is done in (7).

I model the specification for desired hours in the following way:

$$\begin{aligned} \tilde{h} &= h & \text{if } |h - h^* - \omega| \leq \bar{c} \\ \tilde{h} &= h^* \exp(\nu) & \text{if } |h - h^* - \omega| > \bar{c} \\ & & \text{with } \omega \sim N(0, \sigma_\omega^2). \end{aligned} \quad (8)$$

In (8) $\bar{c}$ is a positive parameter (constant across individuals) to be estimated. It is identified by the fact that we have both observations on individuals who are satisfied with their current working hours and on individuals who want to work more or less. The first inequality in (8) describes that observed hours h and optimal hours h^* are not “too far” from each other. The parameter $\bar{c}$ defines the distance and thereby quantifies what is meant

by “too far.” From (8) it follows that the probability that someone is satisfied with the current working hours is

$$\Phi\left(\frac{b - b^* + \bar{c}}{\sigma_\omega}\right) - \Phi\left(\frac{b - b^* - \bar{c}}{\sigma_\omega}\right). \quad (9)$$

The probability density $r(\tilde{b}|w, q)$ of desired hours now consists of the probability (9) for individuals who are satisfied with their current working hours. For those who are not satisfied, and whose desired hours $\tilde{b}$ differ from observed hours b , it consists of one minus the probability in (9), multiplied by the lognormal density for the value of desired hours. The attractiveness of this specification is that it explicitly incorporates the structure of the questionnaire.

D. Likelihood Contributions

In the estimation of the model parameters information both on unemployed and on employed individuals is used. For the unemployed we observe information on unemployment duration t_u . For part of those with a transition into employment accepted wages and hours are observed as well. Information on job tenure t_e of employed individuals is used to estimate the layoff rate, and we also use information on observed wages and hours. Detailed information about the available data is given in the data descriptive section. In this section, I discuss the likelihood contributions for the different types of observations. I will provide likelihood contributions for observations with complete durations sampled according to a flow sample scheme. The extension to right-hand censored observations is straightforward. The likelihood contributions for a stock sample scheme can be obtained by conditioning on backward recurrence times. This procedure is described in appendix B.

Under the given assumptions unemployment duration, conditional on unobserved heterogeneity q , is exponentially distributed with parameter $\theta(q)$, given in (4). After integrating over unobserved heterogeneity the likelihood contribution of an unemployed individual whose after-spell job characteristics are not observed becomes

$$f_u(t_u) = \int_{-\infty}^{\infty} \theta(q) \exp\{-\theta(q)t_u\} g(q, \Sigma) dq, \quad 0 < t_u < \infty. \quad (10)$$

The joint density of observed wages, observed hours, and desired hours (conditional on unobserved heterogeneity) can be derived in three steps. First, the distribution of accepted wages and hours follows from the offer distribution and the condition that accepted wages exceed the (hours dependent) reservation wage ($w > \xi_l(q)$). Next, the joint density function of the observed wage, the accepted wage, observed hours, and desired hours is obtained as the product of the density of the observed wage

conditional on the accepted wage, the joint density of accepted wages and hours, and the density of desired hours. Finally, the latent accepted wage is integrated out by integrating over the range of wages that exceed the reservation wage. Appendix B contains the details of this procedure. It shows that for unemployed individuals whose after-unemployment-spell job characteristics are observed the likelihood contribution can be written as:

$$f(t_u, w_{\text{obs}}, h_l, \tilde{h}) = \int \lambda(q) p_l \exp \{-\theta(q) t_u\} \int_{\xi_l(q)}^{\infty} f_{\text{obs},c}(w_{\text{obs}}|w) f(w) r(\tilde{h}|w, q) dw g(q, \Sigma) dq. \quad (11)$$

In (11) w_{obs} represents the observed wage and w the latent accepted wage offer, while $f_{\text{obs},c}(w_{\text{obs}}|w)$ follows from the density of measurement error in wage rates. Note that the likelihood contribution in (11) is to some extent comparable to that in a competing risk model: individuals can exit unemployment into L different states with different hours levels. Exit to state l occurs with rate $\lambda(v) p_l \bar{F}(\xi_l(q))$.¹⁷ For individuals with no observation on desired hours or model specifications that do not include desired hours, the likelihood contribution can be obtained by integrating (11) over $\tilde{h}$. This results in the following likelihood contribution:

$$f(t_u, w_{\text{obs}}, h_l) = \int \lambda(q) p_l \exp \{-\theta(q) t_u\} \int_{\xi_l(q)}^{\infty} f_{\text{obs},c}(w_{\text{obs}}|w) f(w) dw g(q, \Sigma) dq. \quad (12)$$

For employed individuals for whom we observe job characteristics and desired hours we can determine the likelihood contribution by integrating (11) over duration t_u . This results in the following likelihood contribution:

$$f(w_{\text{obs}}, h_l, \tilde{h}) = \int \frac{\int_{\xi_l(q)}^{\infty} f_{\text{obs},c}(w_{\text{obs}}|w) f(w) p_l r(\tilde{h}|w, q) dw}{\sum_{j=1}^L p_j \bar{F}_j(\xi_j(q))} g(q, \Sigma) dq. \quad (13)$$

The job tenure of employed individuals is exponentially distributed with the layoff rate δ as a parameter. The likelihood contribution due to job tenure is straightforward and needs no further explanation. The likelihood contributions of all the different observations in the sample are multiplied by each other to obtain the likelihood function. Maximization of this

¹⁷ In the appendix it becomes clear that the latter factor cancels against the denominator in (B3).

likelihood function with respect to all the parameters results in the maximum likelihood estimates of these parameters.

In this particular application, the dimension of integrating over unobserved heterogeneity in (10), (11), and (12) is two, and numerical integration is required. Moreover, in model specifications in which the density function of desired hours $r(h|w, q)$ enters, numerical integration over w is required, according to (11). Computational cost is further increased by the need to numerically solve the reservation utility level, which appears in the integrand, from (3). For this reason, we will employ the smooth simulated maximum likelihood method (SSML; see, for instance, Börsch-Supan and Hajivassiliou 1993). We now show how we simulate the likelihood contribution (11). Simulation of (10) and (12) occurs in a comparable way.¹⁸ We draw R replications of unobserved heterogeneity $q_r, r = 1, \dots, R$ from the density $g(q, \Sigma)$, we draw wages w_r from the density $f(w|w > \xi_i(q_r)) = f(w)/\bar{F}(\xi_i(q_r))$, and we simulate the likelihood contribution in (11) by¹⁹

$$\frac{1}{R} \sum_{r=1}^R \lambda(q_r) p_i \exp \{-\theta(q_r) t_u\} f_{\text{obs},c}(w_{\text{obs}} | w_r) r(\tilde{h} | w_r, q_r) \bar{F}(\xi_i(q_r)). \quad (14)$$

Note that (14) is an unbiased simulator of (11). This can easily be seen by taking the expectation of (14) with respect to $(q_r, w_r), r = 1, \dots, R$, taking into account that each (q_r, w_r) is drawn from $g(q, \Sigma)f(w|w > \xi_i(q))$. We use $R = 30$ replications to simulate the integrals. Consistency of the simulated maximum likelihood estimator requires that $R/\sqrt{N} \rightarrow \infty$ as $N \rightarrow \infty$. Börsch-Supan and Hajivassiliou (1993) show that the SSML estimator performs well, even for numbers of replications smaller than 30.

E. On the Identification of Offered and Preferred Hours

Now that we have derived the various likelihood contributions we can address the issue of identification in more detail. Apart from the various variance parameters that enter the econometric specification, we have five sets of structural parameters that enter our model. These are the parameters of the utility function, the hours offer probabilities, the parameters of the wage offer distribution, the parameters of the job offer arrival rate, and the parameters of layoff rate. Without the information on desired hours, there would only be four data series that serve as information on the endogenous variables. These are observed working hours, observed wage rates, unemployment duration, and job tenure. Data on desired working hours are the fifth. First, note that identification of the model parameters in structural job search models always follows from the struc-

¹⁸ Needless to say, for the simulation of (10) we do not need to draw wages.

¹⁹ The replications are such that (q_r, w_r) and (q_s, w_s) are independent for $r \neq s$.

ture of the model. For instance, the hazard rate in (4) depends on all of the model parameters, and the distribution of observed wage rates also depends on all of the model parameters, since it is derived from the distribution of offered wages that exceed the reservation wage. Nevertheless it is clear that in the identification certain sets of parameters have a close link with certain data series. We may say that we can identify the parameters of the job offer arrival rate and the layoff rate by the availability of data on unemployment duration and job tenure, respectively, and the parameters of the wage offer distribution are recovered by the availability of information on wage rates (given the values of the other model parameters). To highlight the identification of preference parameters from hours offer probabilities we write the marginal distribution of observed working hours as²⁰

$$f(h_i) = \int \frac{p_i \bar{F}_i(\xi_i(q))}{\sum_{j=1}^J p_j \bar{F}_j(\xi_j(q))} g(q, \Sigma) dq. \quad (15)$$

The expression in (15) shows us that the probability of observing hours h_i is determined by the hours offer probability p_i and the probability $\bar{F}_i(\xi_i(q))$ of accepting a job with hours h_i . Since we may identify the parameters of the wage offer distribution and the arrival rates from information on wage rates and duration, given the values of the remaining model parameters, what remains to be identified are the parameters of preferences and hours offer probabilities. The marginal density (15) makes clear that changes in offer probabilities and preference parameters both may act to fit the distribution of observed working hours. It is the particular structure of the search model and the nonlinear way in which preference parameters and offer probabilities enter (15) that allow for the parametric identification of the two if only information on working hours is available. It is clear, however, that a change in the preference parameters accompanied by a suitable change in the offer probabilities may lead to little variation in the probability of observed working hours in (15). Thus, a variety of combinations of preference parameters and hours offer probabilities may lead to a “reasonable” (although not exactly the same) fit of the distribution of observed working hours. Once we allow for subjective information on desired hours,²¹ we may “pin down” the preference parameters, and (15) shows that in that case we can identify the hours offer probabilities, conditional on all the other model parameters, from observations on observed working hours. Again, we stress that also with

²⁰ The expression is obtained by integrating (13) over w_{obs} and $\tilde{h}$.

²¹ More sophisticatedly, we may say that our survey data that are described in Sec. III provide information on both hours satisfaction and on desired working hours.

information on desired hours, the structure of the model plays an important role in the identification. This becomes clear once we realize that the marginal distributions of every separate data series depend on all the model parameters, and the structure of the search model determines how the distributions are connected. But with the addition of data on desired working hours, the identification of none of the sets of parameters that we distinguished before exclusively depends on structure of the model.

III. The Data

We use data from the Dutch Socio-Economic Panel (SEP), which is a household panel survey collected by Statistics Netherlands. The households were interviewed twice a year, in April and October. We use data from the October 1985 wave up to and including the April 1989 wave.²² Thus, the total observation period is three and a half years. In each survey wave individuals are asked to report their occupational status for every month in the past 6 months. Selected are male individuals,²³ younger than 65, who reported being unemployed or employed in any month during the observation period. The sample partly has a stock character²⁴ and partly a flow character. For every individual the (complete or incomplete) unemployment or employment duration from the point of sampling has been determined, and for the stock sample observations we add information on backward recurrence times.

Thus, we obtained 573 observations on unemployment duration, of which 297 are observed to end with a transition into employment. Figure 1 plots the Kaplan-Meier estimate of the data on unemployment duration. In the SEP, information on working hours is collected twice a year for individuals employed at the time of the survey. Information on income is collected only in the October wave. For 191 individuals we observe the after-unemployment-spell wage-hours pair. We use subjective data on desired working hours for individuals who are employed at the time of the survey. For 189 of the previously unemployed individuals we observe desired working hours.

We have 4,747 observations on employment duration, which will be used in the estimation of the layoff rate. We observe 252 transitions into

²² In more recent waves (years) of the SEP, information on the individual's labor market state on a month by month basis, which is used to measure unemployment duration, is not collected anymore.

²³ The entire survey contains approximately 5,000 households, aimed to be representative for the Dutch population. Due to low participation rates of females in the Netherlands in the eighties, selecting a sample of unemployed females typically results in a low number of observations.

²⁴ In the estimation, we adjust the likelihood contributions of duration for stock sample observations by conditioning on backward recurrence times as shown in app. B.

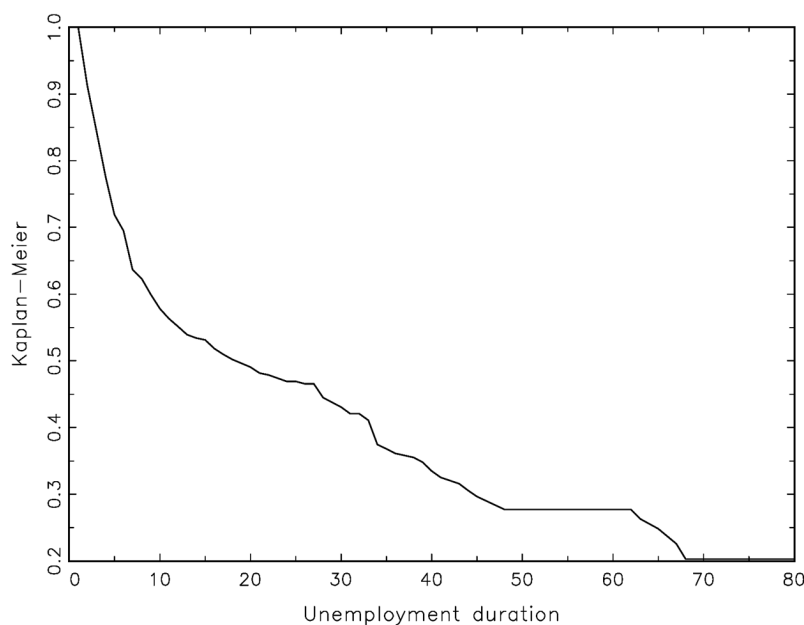


FIG. 1.—Kaplan-Meier plot of data unemployment duration

unemployment. Note that it is not the purpose of the present study to give an extensive and exhaustive explanation of job duration and transitions for the employed. The main reason for adding the observations in job duration is the estimation of the layoff rate that appears in the expression for the reservation utility level (3), which makes it possible to deviate from the standard job search model assumption that unemployed individuals base their evaluation of the gains of search on the view that once a job has been accepted, it will last forever.

For 3,771 of the employed (not including the observations on the previously unemployed, which I have already mentioned before) the wage-hours pair is observed. For 3,216 observations the subjective information on desired working hours is observed. In the analysis, it is assumed that any missing information is missing randomly.

Table 1 shows the sample statistics. The upper panel contains combined information about individuals in the unemployment and the employment spell sample. Information on accepted wages and hours of both types of observation is used in the estimation of the model. The mean number of weekly working hours is 40.6. Survey respondents were asked whether they were satisfied with the working hours in their present job or whether they preferred to work longer or shorter hours, and if they reported preferring to work longer or shorter hours they were asked for their

Table 1
Sample Statistics

Variable	Mean	Standard Deviation
Entire sample ($n = 5,320$):		
Observed working hours (hours/week, $n = 3,962$)	40.6	8.7
Net earnings (guilders/week, $n = 3,962$)	535.3	300.2
Desired hours (hours/week, $n = 3,405$)	39.0	8.1
Difference observed-desired hours ($n = 3,404$)	1.7	6.8
Nonlabor income (guilders/week)	28.9	94.6
Nonlabor income (guilders/week, if positive, $n = 2,087$)	73.6	193.8
Age	35.9	11.3
Family size (persons)	3.2	1.4
	Percentage	
Difference observed-desired hours > 0	25.3	
Difference observed-desired hours $= 0$	68.6	
Difference observed-desired hours < 0	6.1	
Education level	Mode 3	
Dutch nationality	98.1	
Region:		
1. Industrialized west	42.2	
2. East	24.0	
3. South	23.2	
4. Agricultural	10.6	
Type of education:		
1. Technical	41.3	
2. Economic/administrative	15.6	
3. No specialization	29.0	
4. Services	14.1	
	Mean	Standard Deviation
Unemployment spells ($n = 573$):		
Benefits (guilders/week)	162.8	166.4
Benefits (guilders/week, if positive, $n = 317$)	294.3	106.3
Nonlabor income (guilders/week)	20.0	53.7
Nonlabor income (guilders/week, if positive, $n = 163$)	70.1	81.4
Unemployment duration (months)	14.2	15.8
Age	34.2	11.7
Family size (persons)	2.9	1.4
	Percentage	
Education level	Mode 2	
Dutch nationality	95.6	
Married	50.4	
Region:		
1. Industrialized west	30.9	
2. East	31.4	
3. South	24.8	
4. Agricultural	12.9	
Type of education:		
1. Technical	37.5	
2. Economic/administrative	10.3	
3. No specialization	39.6	
4. Services	12.6	

Table 1 (Continued)

Variable	Mean	Standard Deviation
Employment spells ($n = 4,747$):		
Nonlabor income (guilders/week)	30.0	98.4
Nonlabor income (guilders/week, if positive, $n = 1,924$)	73.9	143.6
Employment duration	87.5	109.2
Age	36.1	11.2
Family size (persons)	3.3	1.3
	Percentage	
Education level	Mode 3	
Dutch nationality	98.4	
Region:		
1. Industrialized west	43.6	
2. East	23.1	
3. South	23.0	
4. Agricultural	10.3	
Type of education:		
1. Technical	41.8	
2. Economic/administrative	16.2	
3. No specialization	27.7	
4. Services	14.3	

desired number of working hours in their present job. Given the phrasing of the question, it is assumed that the answers of the survey respondents are conditional on remaining job characteristics, including the hourly wage rate. The model specifications for desired hours will allow for measurement error in desired hours. Note that for individuals who report being satisfied with current working hours we put desired hours equal to observed hours. For individuals who report not being satisfied with current working hours observed hours are different from desired hours. In this way the observed series of desired working hours was defined. In Section II.C two alternatives are presented for modeling the link between the desired working hours and the optimal working hours defined by the model.

The mean number of desired working hours is 39.0, which is slightly lower than the actual number of working hours. Among the individuals whose actual and desired working hours are both observed, 68.6% are satisfied with the number of hours worked at the present job, 25.3% report being overemployed, and 6.1% would like to work more.²⁵ Although there is a sizable minority of over 30% of the respondents that report not being satisfied with the current number of working hours, the empirical distribution of desired working hours has properties that are more similar to the empirical distribution of observed working hours than

²⁵ Tables 4 and 5 contain more details about the empirical distribution of observed and desired working hours.

to a distribution that may be generated by optimal working hours that stem from the neoclassical model of labor supply. For example, the distribution of desired working hours shows a peak at a working week of 40 hours, although the peak is 12% points lower than the peak of the empirical distribution of observed working hours. It is likely that when being asked to report satisfaction with current working hours, individuals report being satisfied as long as their optimal working hours are “not too far away from,” instead of “equal to,” their current working hours. Thus, we may incorporate that “satisfaction” is different from “optimality.” In Section II.C I explicitly showed how to model the relation between optimal working hours and desired working hours on the basis of this idea.

The average age is about 36, and the mean family size is 3.2. Four levels of education are distinguished, where level 1 is the lowest and level 4 the highest. The Netherlands is divided into four regions. Region 1 is the most strongly industrialized part of the Netherlands, which includes the larger cities. Region 4 is the least industrialized part of the Netherlands with a relatively low population density and a sizable agricultural sector. Region 3 is the south of the Netherlands, which contains some large companies and agricultural industry. In region 2 (the east) there is a mix of industry and agriculture. Apart from having information about the level of education, information is available about the type or sector of education. Sector 1 is the technical sector, which includes chemistry, physics, mathematics, and biology; sector 2 includes the economic and administrative directions; sector 3 is general education; and sector 4 includes services.

The mean unemployment duration in the sample of 573 unemployment spells is 14 months, whereas the median (not shown in the table) is 6 months. The average age of the individuals in the sample with unemployment spells is about 34, whereas the mean family size is 2.9. The mean employment duration in the sample of 4,747 employment spells is 87.5, about 7 years, whereas the median employment duration (not shown in the table) is 3 years.

IV. Results

A. Parameter Estimates

The parameter estimates of the three model variants are presented in table 2, which shows the estimates of the utility parameters. As is typical in the labor supply literature, information related to household composition is included: the household family size and the marital status of the respondent appear as covariates in the utility function. Job search models without working hours usually contain only the arrival rate and the wage offer distribution, so information on household characteristics is usually not included. The identification in the base model of structural parameters

Table 2
Parameters of the Utility Function and the Distribution of Desired Hours

Variable/ Parameter	The Base Model		Optimal Desired Hours Model		Hours Satisfaction Model	
	Estimate	Standard Error	Estimate	Standard Error	Estimate	Standard Error
β	-.044**	.002	-.023**	.001	-.042**	.001
γ	.79**	.05	.63**	.03	.52**	.02
Intercept	31.6**	1.2	24.7**	.5	24.4**	.5
Family size	4.6**	.7	2.7**	.3	2.4**	.3
Married	3.5**	.8	2.5**	.3	2.6**	.4
Age ≤ 25	.8	1.2	4.5**	.4	7.0**	.6
25 < Age ≤ 35	4.4**	1.0	2.5**	.3	3.1**	.3
35 < Age ≤ 45	4.3**	1.2	1.4**	.4	2.2**	.5
σ_e	13.9**	.3	8.3**	.1	9.8**	.1
σ_v			.20**	.00	.14**	.001
$\bar{c}$					16.3**	.6
σ_ω					12.9**	.6

* Indicates significance at the 10% level.

** Indicates significance at the 5% level.

of the utility function like β and γ is achieved because they measure the impact of income on the unemployment duration by their impact on the reservation wage and the acceptance probability. Note that for the identification of the parameters of the household characteristics in the utility function the exclusion restriction that household characteristics only enter utility plays an important role. Age dummies are included in both the utility function and the arrival rate. As a consequence, in the base model information on unemployment duration is used to identify both the effect of age on preferences and the influence of age on the arrival rate, and it is clear that here identification of the age effects leans on the imposed structure and functional form. Once information on desired hours is included, separate identification of the effects of age on utility and on the arrival rate is possible.²⁶

Note that the parameters β and γ are largest in the base model. By (6), this implies larger effects of nonlabor income and the wage rate on optimal working hours compared to the estimates obtained with the remaining two model variants. If we compare the “optimal desired hours model” (desired hours specified according to [7]) with the “hours satisfaction model” (desired hours specified according to [8]), we see that the optimal working hours according to the hours satisfaction model show a higher

²⁶ Also in the specification in which desired working hours are used, household characteristics are not included in the job offer arrival rate, although identification is not a problem there. I use only characteristics like age and education that often are assumed to be directly related to a worker's productivity. A good reason to include household characteristics is that the search intensity of the individual may be affected by household composition, if they affect the cost of search.

sensitivity with respect to the nonlabor income than the optimal working hours implied by the optimal desired hours model. A possible explanation for this is that the optimal desired hours model imposes a tight relation between the optimal hours and the observed desired hours and therefore allows for less variation in optimal hours. The three model variants show a significant impact of family size on preferences: a larger family size is associated with stronger preferences in favor of working. The same holds for marital status: married men like to work more hours. The effects of family size and marital status are smaller if information on desired hours is included in the estimation of the model. The base model shows an age pattern different from the two models that include desired hours. Both variants with desired hours show that optimal hours decrease monotonically with age, whereas there is an inverted U-shaped pattern for the base variant. Apart from the difference in the age pattern, there is also a difference in the magnitude of the age effects: for the groups aged between 25 and 45, the model variants with desired hours show a smaller effect of age. Another reflection of the identifiability of preferences for working hours is given by the size of the estimated variance σ_ϵ of the random preferences. It is largest for the base model, which indicates that the base model assigns a relatively large part of variation in preferences to unobserved heterogeneity. It is smallest for the optimal desired hours model that imposes a close link between the observations on desired hours and optimal working hours. In the hours satisfaction this link between desired hours and optimal hours is much weaker. Yet the estimated variance of random preferences according to the hours satisfaction model is only slightly larger than the estimate according to the optimal desired hours model. This indicates that the information on desired hours still adds to the identifiability of preferences for working hours in the hours satisfaction model, in spite of the weaker link between desired hours and optimal hours. Table 2 also shows the parameters $\bar{c}$ and σ_ω that are specific to the hours satisfaction model. We see that the estimate of the gap $\bar{c}$ is 16.3 and its standard deviation σ_ω is 12.9. Thus, the estimated gap and its standard error allow for a large difference and variation between optimal hours and hours that the respondent reports being satisfied with.

Table 3 shows the parameter estimates of the job offer arrival rate. The base model shows a U-shaped impact of age on the arrival rate. Both models with desired hours show that only the youngest age group has a significant larger arrival rate. The earlier discussion on identification showed that identification of the base model is achieved by functional form: in the base model the larger impact of age on preferences in favor of working that we found for the middle two age groups apparently has to be compensated by a smaller impact of age for these groups on the job offer arrival rate, compared to the model with desired hours, in order to obtain a comparable impact on unemployment duration. This result

Table 3
Parameters of the Job Offer Arrival Rate

Variable/Parameter	The Base Model		Optimal Desired Hours Model		Hours Satisfaction Model	
	Estimate	Standard Error	Estimate	Standard Error	Estimate	Standard Error
Intercept	-1.16**	.27	.83	.72	2.8**	.6
Age ≤ 25	-.24	.16	.54**	.28	.41*	.24
25 < Age ≤ 35	-.29**	.13	.17	.21	.10	.17
35 < Age ≤ 45	-.17*	.10	.05	.17	.05	.14
Educ1	7.8**	1.0	5.3**	1.3	6.2**	1.6
Educ2	4.6**	.4	3.2**	.6	4.3**	.8
Educ3	4.6**	.2	3.9**	.7	7.0**	.7
Region:						
West	.17	.11	.30*	.15	.25**	.12
East	-.20*	.11	-.18	.17	-.16	.13
South	-.02	.12	.02	.18	-.01	.14
Type of education:						
Technical	2.25**	.36	8.7**	1.8	4.5**	.8
Economic/administrative	2.72**	.38	4.0**	1.3	2.3**	.9
No specialization	-.66**	.22	-.29	.61	-.78	.55
σ_λ	.26**	.06	.065	.084	.010	.07

* Indicates significance at the 10% level.

** Indicates significance at the 5% level.

shows the value of using subjective information on desired working hours for the identification of the way in which demographics enter the model. All of the three model variants show positive and significant education dummies, possibly due to more complicated and lengthy application and hiring procedures for the higher educated. For the sectoral dummies the models yield similar results. The models assign the highest arrival rates to individuals with a technical and an economic/administrative type of education, although the base model assigns the highest job offer arrival rate to the economic/administrative sector, whereas the models with desired hours assign the highest arrival rate to the technical sector. Finally, the variance of the unobserved heterogeneity in the arrival rate is higher in the base model than in the model variants with desired hours. Note that the estimates in both the model variants with desired hours are not significant.²⁷

Table 4 shows the parameter estimates of the layoff rate. The differences between the model variants are not very large. They all imply that the layoff rate decreases with age and that individuals with the two lowest levels of education have the highest layoff rate. Individuals in the service

²⁷ We experimented with imposing correlation between the random preferences ϵ and the unobserved heterogeneity v . In the base model, the correlation was insignificant. In the model with desired hours, it was hard to identify the parameter, which is understandable once we see the small and insignificant variance in table 3 that is obtained if we do not impose correlation.

Table 4
Parameters of the Layoff Rate

Variable	The Base Model		Optimal Desired Hours Model		Hours Satisfaction Model	
	Estimate	Standard Error	Estimate	Standard Error	Estimate	Standard Error
Intercept	-8.0**	.3	-8.0**	.3	-8.3**	.1
Age ≤ 25	2.6**	.2	2.5**	.2	2.5**	.1
25 < Age ≤ 35	1.4**	.2	1.4**	.2	1.5**	.1
35 < Age ≤ 45	.57**	.23	.64**	.23	.74**	.22
Educ1	.08	.24	.54**	.25	.59**	.11
Educ2	.27	.20	.57**	.21	.56**	.10
Educ3	-.63**	.21	-.39*	.21	-.42**	.10
Region:						
West	-.24	.19	-.22	.19	.004	.12
East	.22	.19	.12	.19	.35**	.09
South	.12	.19	.11	.19	.32*	.17
Type of education:						
Technical	-.43**	.17	-.58**	.17	-.55**	.16
Economic/administrative	-.43*	.23	-.69**	.24	-.66**	.23
No specialization	-.26	.18	-.61**	.20	-.65**	.19

* Indicates significance at the 10% level.

** Indicates significance at the 5% level.

sector have the highest layoff rate. Region does not affect the layoff rate. The largest differences are found for education: the models estimated with information on desired hours show that workers with one of the lower two education levels have a significantly higher layoff rate, which is plausible. For the base model, the effect for these education levels is positive but not significant.

The parameter estimates of the wage offer distribution in table 5 show that offered wages increase with the level of education and show an inverted U-shaped pattern in age (with peaks at the ages of 44 and 49). For all the model variants, the offered wages are lowest for individuals with a technical and economic type of education.

Finally, table 6 shows the parameter estimates of the hours offer distribution. Note that some restrictions on the probabilities of certain hours categories have been placed due to low numbers of observations in certain cells. From the discussion on identification in Section II it followed that without the use of information on desired hours, it is impossible to identify offered hours from preferred hours. For instance, if we want to know whether the observation that fewer part-time jobs than full-time jobs are observed is due to the fact that employers do not offer these jobs or whether workers do not accept them, model estimates obtained with information on observed hours only may generate very unreliable results. All the model variants show a peak in the distribution of offered hours at 40 hours a week. Moreover, the base model assigns higher offer probabilities to weekly numbers of working hours below 40, whereas the

Table 5
Parameters of the Wage Offer Distribution

Variable/Parameter	The Base Model		Optimal Desired Hours Model		Hours Satisfaction Model	
	Estimate	Standard Error	Estimate	Standard Error	Estimate	Standard Error
Intercept	1.3**	.1	.83**	.18	.17	.14
log(Age/16)	1.8**	.1	1.6**	.1	1.48**	.10
log(Age/16) ²	-.9**	.1	-.8**	.1	-.69**	.06
Educ1	-1.61**	.13	-.98**	.15	-1.24**	.22
Educ2	-1.03**	.06	-.62**	.07	-.88**	.11
Educ3	-.99**	.04	-.65**	.08	-1.17**	.10
Type of education:						
Technical	-.38**	.06	-.96**	.19	-.62**	.10
Economic/administrative	-.39**	.06	-.43**	.16	-.28**	.12
No specialization	.11**	.04	.04	.09	.12	.09
τ	.52**	.01	.48**	.03	.63**	.02
σ_m	.23**	.004	.31**	.002	.30**	.002

* Indicates significance at the 10% level.

** Indicates significance at the 5% level.

models with desired hours assign higher offer probabilities to working hours above 40. As we will see later on, this is accompanied by a reversed pattern in optimal hours: the base model simulates higher frequencies of desired hours above 40 and shows that individuals will be underemployed on average. The latter is in contrast with the survey information on satisfaction with working hours.

B. Benefit Elasticities and Reservation Wages

To compare the implications of the models I computed the elasticity of the expected unemployment duration with respect to the benefit level for a given set of individual characteristics. For benefits, the nonlabor income, family size, and the age of the individual I took the sample means for the unemployed from table 1. For the level of education, the sector of education, the region, and marital status I took the lowest level of education, no specialization, the western region, and married. We averaged over 13,000 replications from the distribution of unobserved heterogeneity.

To see how reservation wage rates vary with hours, I computed the reservation wages for every hours category for the given type of individual. Again, I averaged over unobserved heterogeneity. Recall from Section II that due to variation in preferences with respect to working hours the reservation wage rate is a function of working hours that reaches a minimum in optimal hours. Note that if preferences would not depend on working hours, reservation income is constant and the reservation wage rate monotonically decreases with hours, the latter because working,

Table 6
Parameters of the Hours Offer Distribution

Hours $h_l =$	The Base Model		Optimal Desired Hours Model		Hours Satisfaction Model	
	Estimate $P(h = h_l)$	Standard Error	Estimate $P(h = h_l)$	Standard Error	Estimate $P(h = h_l)$	Standard Error
4, 8, 12, 16	.032**	.006	.032**	.006	.012**	.002
20	.067**	.008	.043**	.005	.025**	.003
24	.030**	.004	.016**	.002	.012**	.002
28	.019**	.003	.010**	.002	.0094**	.0014
32	.039**	.004	.022**	.002	.025**	.002
36	.034**	.003	.024**	.002	.027**	.002
40	.55**	.02	.47**	.02	.57**	.01
44	.047**	.003	.052**	.004	.062**	.004
48	.012**	.001	.018**	.002	.020**	.002
52	.031**	.003	.063**	.005	.068**	.005
56	.0083**	.0012	.024**	.003	.024**	.003
60	.014**	.002	.053**	.006	.050**	.005
64, 68	.0021**	.0004	.012**	.002	.0093**	.0015
72	.0055**	.0010	.048**	.009	.033**	.005
76, 80, 84	.0029**	.0005	.0028**	.0002	.0030**	.0001

NOTE.— $p_l = P(h = l \times 4)$ or, more precisely, $P(l - 2 \leq h < l + 2)$.

* Indicates significance at the 10% level.

** Indicates significance at the 5% level.

for example, twice as many hours requires the wage rate to be twice as low to keep labor income constant.

Figure 2 plots reservation wage rates as a function of hours in the range of 16–84. The graph contains the reservation wage rates for the base model as well as for the two model variants estimated with desired hours. For reasons of comparison, we also plotted the wage rates that correspond with a constant reservation income.²⁸ It is clear that the reservation wage rates of the base model show a pattern that is closer to the constant reservation income case than the models estimated with desired hours. Note that the reservation wage as a function of hours is much flatter for the base model than for the optimal desired hours model: for the base model, there is not much variation in the reservation wage for a working week of 40 hours or higher, implying that preferences are not very sensitive with respect to these hours levels. In other words, we cannot really identify the individual preferences for different working hours from the estimation results for the base model. For the base model, the minimum reservation wage is reached for hours levels of 48–60, implying that this is the optimal number of working hours for an individual of the given type. This would suggest that there is underemployment. The optimal desired hours model reaches its minimum at 36–40 hours a week. Note

²⁸ The value of the constant reservation income is equal to the reservation income of the base model at 40 hours a week. The plot of the constant reservation income wage rates at this particular value is shown only for reasons of exposition.

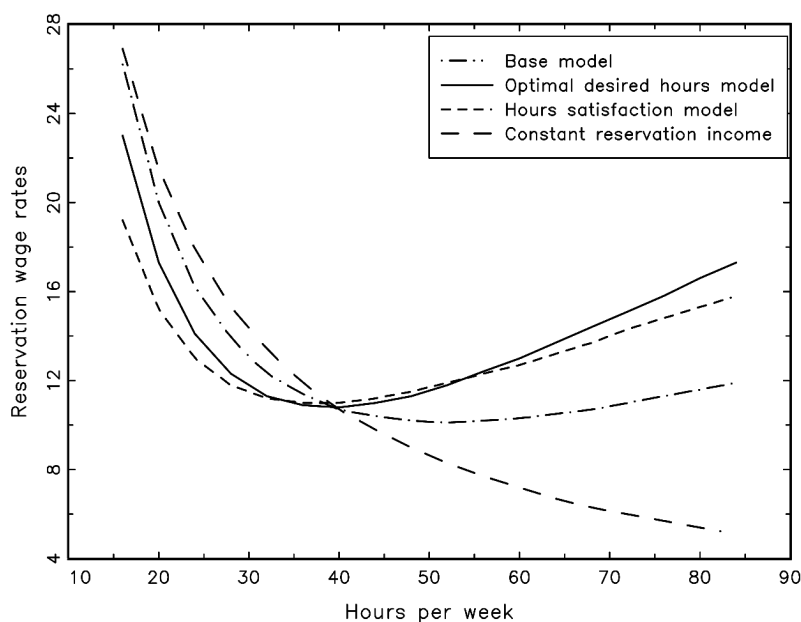


FIG. 2.—Reservation wage rate as a function of hours

that for this model, reservation wages are lowest in the range of 32–52 hours a week. The hours satisfaction model also reaches a minimum at 36–40 hours a week. Compared to the first two model variants, the hours satisfaction model shows lower reservation wage rates in low numbers of working hours, implying a higher acceptability of low working hours. Within the range of hours from 30 to 50 the reservation wage rates of the optimal desired hours model and the hours satisfaction model look similar, and both show an obvious minimum. The possibility of locating a minimum in the graph is again an indication that the hours satisfaction model is able to identify preferences over working hours.

Table 7 shows the elasticities. The elasticity that measures the impact of the benefit level on the expected unemployment duration is 0.35 for both the base model and the optimal desired hours model. The hours satisfaction model displays with an elasticity of 0.50 a larger sensitivity of the expected unemployment duration with respect to the benefit level. Table 7 also shows the elasticities of the reservation wage rates with respect to the benefit level. It shows slightly larger elasticities for the base model than for the optimal desired hours model. For example, at 40 hours a week the elasticity of the reservation wage with respect to the benefit level is 0.08 for the base model and 0.06 for the desired hours model,

Table 7
Elasticities of Expected Unemployment Duration and Reservation Wages

	The Base Model	Optimal Desired Hours Model	Hours Satisfaction Model
Elasticity of Expected Unemployment Duration with Respect to Benefit Level			
Hours	.35 (.02)	.35 (.05)	.50 (.04)
Elasticity of Reservation Wage Rate with Respect to Benefit Level			
4	.02 (.002)	.02 (.002)	.05 (.002)
8	.03 (.003)	.03 (.002)	.07 (.003)
12	.04 (.004)	.04 (.003)	.09 (.003)
16	.05 (.004)	.05 (.004)	.11 (.004)
20	.06 (.005)	.05 (.004)	.12 (.004)
24	.06 (.005)	.06 (.004)	.13 (.004)
28	.07 (.005)	.06 (.005)	.14 (.005)
32	.07 (.006)	.06 (.005)	.14 (.005)
36	.08 (.006)	.06 (.004)	.14 (.004)
40	.08 (.006)	.06 (.004)	.14 (.004)
44	.08 (.006)	.06 (.004)	.13 (.004)
48	.08 (.006)	.05 (.004)	.13 (.004)
52	.08 (.005)	.05 (.004)	.12 (.004)
56	.08 (.005)	.05 (.003)	.11 (.003)
60	.07 (.005)	.04 (.003)	.11 (.003)
64	.07 (.005)	.04 (.003)	.10 (.003)
68	.07 (.005)	.04 (.003)	.10 (.003)
72	.07 (.005)	.04 (.003)	.09 (.003)
76	.06 (.005)	.03 (.002)	.09 (.003)
80	.06 (.004)	.03 (.002)	.09 (.003)
84	.06 (.004)	.03 (.002)	.08 (.002)

NOTE.—Standard errors of elasticities are in parentheses.

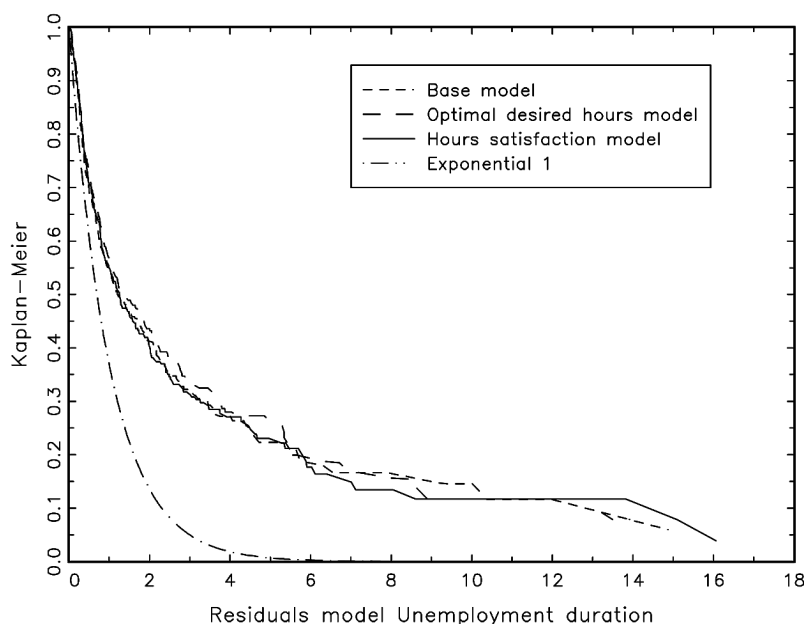


FIG. 3.—Kaplan-Meier plot of the residuals for unemployment duration

whereas the values are not in each others' 95% confidence interval. The hours satisfaction model shows the largest elasticities.

C. Residual Analysis

To analyze how the model fits the data on unemployment duration we plotted the residuals. Note that the three model variants are all stationary structural duration models, which do not account for possible duration dependence of the hazard. The models, though, do incorporate various sources of unobserved heterogeneity. The residuals of the models are plotted in figure 3. The dashed-dotted line in the figure is the survivor function according to an exponential distribution with parameter 1. If the models are correctly specified, the residuals should be distributed according to this distribution. Although it is clear that the models would not survive a formal test, the results are not unreasonable for a stationary structural duration model. The models do reasonably well for shorter durations but fail to explain some of the larger durations. The residuals reveal evidence of neglected negative duration dependence of the hazard. The fit of the base model is comparable to the model variants estimated with desired hours, even though the use of additional data on desired hours imposes additional restrictions on the parameters of the utility function.

The negative duration dependence, indicated by the residual analysis, is consistent with many reduced-form empirical studies, most of which find negative duration dependence (see, e.g., Lancaster 1979; Lynch 1985; Blau and Robins 1986). Any evidence of duration dependence in reduced-form models for unemployment duration is generally interpreted as evidence against stationarity.²⁹ Blau and Robins (1986) also show that negative duration dependence of the hazard rate can be the net result of arrival rates and reservation wages that both decline with unemployment duration.³⁰ Bloemen (2005), in a structural stationary model of job search with endogenous search intensity, also finds evidence of neglected stationarity and negative duration dependence. He shows with a separate residual analysis for forward and backward recurrence times that the exponential model (implied by the stationarity assumption) is notably unable to explain the longer unemployment durations.

D. The Distribution of Working Hours

For the different model specifications the distribution of working hours has been simulated. For every individual with observed working hours in the sample I computed the distribution of working hours, implied by the model, conditional on unobserved heterogeneity q , for 30 replications of q and averaged over these 30 replications and individuals to obtain the simulated hours frequencies. I also simulated the optimal number of working hours, implied by the labor supply function (6) by computing the optimal number of working hours for every individual in the sample with observed desired working hours and for 30 replications of random preferences. The results of the simulations are shown in tables 8 and 9. The observed working hours show a peak of 67% around a working week of 40 hours. The simulated hours distributions of the models approximate the empirical distribution of observed hours closely. Given the flexible specification of the hours offer distribution, this is no surprise. Adding the information on desired hours teaches us more about the underlying preference structure with respect to working hours. The frequencies of optimal hours according to the base model are flat. The base model does succeed in assigning low frequencies to low weekly numbers of working hours, although the frequencies are higher than for the distribution of observed desired working hours. The base model fails to track the location of desired working hours and also assigns too much probability mass to numbers of weekly working hours that exceed 40. The optimal desired

²⁹ In structural empirical search models it is much more complicated to include nonstationarity and duration dependence; see Van den Berg (1990).

³⁰ Note that declining reservation wages in isolation lead to an increasing hazard. Kiefer and Neuman (1979) also provide evidence of reservation wages that decline significantly with duration.

Table 8
The Distribution of Desired, Optimal, and Simulated Desired Hours

Hours Category $h =$	Observed Desired Hours	Optimal, according to Base Model	Optimal, Optimal Desired Hours Model	Optimal, Hours Satisfaction Model	Simulated Desired Hours, Optimal Desired Hours Model	Simulated Desired Hours, Hours Satisfaction Model
4	.001	.008	.003	.011	.003	.011
8	.002	.006	.005	.013	.007	.012
12	.002	.011	.014	.028	.020	.025
16	.004	.019	.034	.051	.045	.041
20	.023	.030	.064	.082	.078	.062
24	.017	.044	.103	.117	.113	.065
28	.010	.058	.143	.144	.134	.064
32	.112	.077	.165	.154	.139	.066
36	.117	.090	.160	.140	.124	.057
40	.548	.101	.129	.111	.104	.415
44	.055	.106	.088	.073	.077	.060
48	.017	.104	.051	.042	.055	.026
52	.047	.092	.024	.021	.037	.043
56	.012	.078	.010	.009	.024	.016
60	.017	.060	.004	.003	.015	.020
64	.004	.044	.001	.001	.009	.004
68	.001	.031	.001	.000	.006	.003
72	.005	.019	.000	.000	.003	.007
76	.001	.011	.000	.000	.002	.001
80	.003	.006	.000	.000	.001	.001
84	.002	.007	.001	.000	.002	.000

NOTE.—Hours category with hours = h : $l-2 \leq h < l+2$; $l = 4$: lower bound 0, $l = 84$: upper bound ∞ .

hours model does better in the sense that the model places most of the probability mass of desired working hours in the area of 32–44 hours a week. The model does not succeed in predicting the peak and the asymmetry in desired working hours. However, the model with desired hours succeeds in predicting that there are more overemployed than underemployed individuals. For the hours satisfaction model table 8 shows that by the specification in (8) we manage to predict a sizable peak in the distribution of desired hours at 40 hours a week. In fact the model explains over 75% of the peak in the data on desired hours. Overall the pattern in the simulated desired hours is somewhat flatter than the pattern in the observed desired hours. Nevertheless, compared to the simulated desired hours of the optimal desired hours model, in which, by (7), the only difference between optimal hours and desired hours is a lognormally distributed measurement error, it is a big improvement. Also note that the hours satisfaction model predicts overemployment better than the optimal desired hours model.

As a final check, we have computed a goodness of fit test statistic for the distribution of observed hours. Table 9 shows that, with the naked

Table 9
The Distribution of Observed and Simulated Observed Hours

Hours Category $h =$	Observed Hours Frequencies	Simulated with Base Model	Simulated, Optimal Desired Hours Model	Simulated with Hours Satisfaction Model
4	.001	.001	.001	.001
8	.002	.002	.001	.003
12	.004	.004	.005	.006
16	.005	.006	.012	.010
20	.020	.019	.034	.026
24	.013	.012	.018	.014
28	.011	.011	.014	.012
32	.031	.030	.033	.030
36	.034	.033	.036	.032
40	.670	.665	.647	.659
44	.067	.067	.065	.067
48	.019	.019	.019	.020
52	.054	.055	.054	.057
56	.016	.016	.016	.017
60	.027	.027	.027	.028
64	.007	.004	.005	.004
68	.001	.004	.003	.003
72	.010	.010	.010	.009
76	.001	.005	.0004	.001
80	.005	.005	.0003	.0004
84	.002	.004	.0002	.0003

NOTE.—Hours category with hours = l : $l-2 \leq h < l+2$; $l=4$: lower bound 0, $l=84$: upper bound ∞ .

eye, the distributions of observed working hours obtained with the three different models look very similar. However, a formal goodness of fit test may reveal differences in the fit of the different models. We use the χ^2 goodness of fit test proposed by Heckman (1984).³¹ We have grouped observed working hours into 15 classes. The classes coincide with the hours classes that have been imposed in the estimation of the hours offer distribution (see table 6).³² The test statistic is distributed according to the χ^2 distribution with 14 ($= 15 - 1$) degrees of freedom. The critical value at the 5% level is 23.7. The outcomes of the test statistic for the base model, the optimal desired hours model, and the hours satisfaction model are, respectively, 17.9 (p -value 0.21), 264, and 225. The base model

³¹ Andrews (1988) presents a generalization of the χ^2 test, which includes the test by Heckman (1984) as a special case. We have followed Heckman (1984) in computing the estimate of the covariance matrix that enters the computation of the test statistic. See Heckman (1984) for details.

³² The null hypothesis is that the probability of observing working hours in a certain class follows from the parametric model. More specifically, we test whether the probability distribution of observed working hours according to the parametric model is equal to the nonparametric empirical distribution of observed working hours. Consequently, the alternative hypothesis is general and not restricted to any specific parametric form.

passes the goodness of fit test for observed working hours. The test results for the optimal desired hours model and the hours satisfaction model show that the parametric model specification places restrictions on the probability distribution of observed working hours that make it different from the nonparametric empirical distribution of observed working hours. This means that the latter two models are not sufficiently able to predict the correct hours category at the individual level, even though the distribution of the sample as a whole seems to fit well.

The results show that the base model fits the observed distribution of working hours well, both by inspection of the simulated distribution function and according to a goodness of fit test. In the estimation of the base model the information on desired hours is not incorporated. Therefore the model is overparameterized: supply side parameters (preferences) and demand side parameters (hours offer distribution) are both retrieved using observed hours only.³³ This explains why the base model fits observed working hours better than the alternatives. A comparison of the computed optimal working hours according to the base model with the data on desired working hours, however, shows that the base model is not able to reproduce some basic features of the data, like the location of the distribution of desired working hours and the working hours at which the reservation wage is lowest. This reveals the paradox between fitting the distribution of observed working hours, on the one hand, but not being able to explain the underlying behavior, on the other. The optimal desired hours model and the hours satisfaction model exploit the information on desired working hours in the estimation. These models succeed in replicating the location of desired working hours and the data feature that the majority of the observations are overemployed. The hours satisfaction model in turn is better able to explain the peak in the data on desired working hours. Overparameterization is not present in the latter two models, so with virtually the same set of parameters as in the base model we now have to fit both the distribution of observed working hours and the distribution of desired working hours.³⁴

Apart from the stationarity assumption in the search model, and within the structure of the present model, we distinguish two main possible causes behind the result of the goodness of fit test. First, the functional form of the utility function may be too restrictive. In Section II.B we have already discussed this issue. Second, there may have been included insufficient observed heterogeneity to be able to fit the model at the individual level. To improve the fit we may either introduce observed characteristics in $\bar{c}$ in (8) or introduce observed characteristics in the hours

³³ This has been spelled out in Sec. II.E, using the marginal distribution of observed working hours (15).

³⁴ See again Sec. II.E and (15).

offer probabilities. About observed characteristics in $\bar{c}$ it should be noted that these observed characteristics will neither have the interpretation of taste shifters (which are included in the utility function) nor have the interpretation of demand side factors (which are determined by the hours offer probabilities). Therefore, observed characteristics in $\bar{c}$ may be hard to interpret in terms of economic theory. Moreover, the impact on the fit of the distribution of observed working hours is indirect. In the first place it improves the fit of the distribution of desired hours. Making the hours offer probabilities dependent on observed characteristics is likely to be an easy way to improve the fit of the model.³⁵ A disadvantage is that we may not be sure whether we really will be estimating the impact of observed characteristics on the hours offers, and instead we may simply be correcting for anything that cannot be explained by the form of the preferences.³⁶

E. Policy Simulations

Understanding the role of hours restrictions is potentially very important to assess the effects of some labor market reforms that have been at the center of the political debate in recent years. For all the policy simulations described in this section, we used the hours satisfaction model as our preferred model. We address various policy questions that involve the combination of benefits, unemployment duration, hours offer restrictions, and preferences. It should be noted that any policy change may lead to a change in distribution of job offers or wage offers. In the policy simulations, we take these distributions as given and we do not consider possible general equilibrium effects due to changes in firm behavior.

First, we consider the impact of a cut in benefits on the distribution of working hours. In the political debate a cut in unemployment benefits is advocated because it increases the incentives of the unemployed to accept a job, that is, reservation wages decrease and the unemployment duration falls. Opponents of a decrease in benefits argue that the faster acceptance of jobs comes at the cost of the match between job and worker. In the context of the choice of working hours this may mean that workers who would like to work part-time (full-time) accept a full-time (part-time) job more easily if the benefit level is lower. Since a fall in the benefit level decreases the value of unemployment, reservation wages for jobs with any hours level will fall (see table 7). But since reservation wage

³⁵ Section IV.A, for instance, showed different age patterns for the base model, on the one hand, and the optimal desired hours model and the hours satisfaction model, on the other.

³⁶ In Sec. II.A it was noted that the distribution of offered working hours may largely be determined by institutional level, and therefore the “true” influence of individual heterogeneity may be limited.

Table 10
Impact on Accepted Hours of a 5% Decrease in Benefits

Hours Category $h =$	Hours Distribution (Percentages) at Average Benefits	Hours Distribution (Percentages) at Decreased Benefits
4	.0	.0
8	.0	.0
12	.2	.2
16	.5	.7
20	1.7	2.2
24	1.1	1.3
28	1.0	1.2
32	2.9	3.0
36	3.3	3.5
40	68.1	66.8
44	7.0	6.9
48	2.1	2.0
52	5.9	5.8
56	1.7	1.7
60	2.9	2.9
64	.4	.5
68	.3	.4
72	.9	1.0
76	.1	.0
80	.0	.0
84	.0	.0

NOTE.—Hours category with hours = h : $h-2 \leq h < h+2$; $h=4$: lower bound 0, $h=84$: upper bound ∞ .

rates for different hours levels are different, the distribution of accepted hours can be affected. To quantify the impact of a fall in the benefit level, we first simulated the distribution of accepted working hours for an individual with the average characteristics of the unemployed in the sample (see table 1). We have drawn 13,000 replications from the distribution of heterogeneity in preferences and the arrival rate to obtain the distribution of accepted working hours. This procedure was repeated while benefit levels were decreased by 5%. The hours distributions (in percentage points) for the different benefit levels are shown in table 10. Two observations can be made. Overall the impact of the 5% decrease in benefits on the distribution of accepted working hours is limited. There is, however, a small shift toward part-time hours. For instance, the frequency of working full-time hours (40 hours a week) has decreased by 1.3 percentage points, whereas the frequency of working half time (20 hours a week) has increased by 0.5 percentage points. If benefits fall it becomes more attractive to accept jobs with less than full-time hours which generate a lower (weekly) income. Note that the 5% decrease in benefits is apparently not large enough to find a shift toward accepting jobs with more than 40 hours a week, even though (a priori) lower benefits may induce someone unemployed to accept a job with an unfavorably high number of working hours.

Next, there is an ongoing discussion about the length of the (standard) working week. From the data section it became clear that most jobs have hours levels concentrated around 40 hours a week but that a considerable fraction of the workers indicates that they prefer to work fewer hours. The government may influence the cost of employers to employ part-time workers or the length of the standard working week may be decreased. These changes influence the hours offer distribution. I simulated two different changes in the hours offer distribution. The first is a straightforward change in the length of the working week. I exchanged the offer probabilities of working 40 hours and working 36 hours. In terms of table 6, the offer probability of 36 hours becomes 0.57 and the probability of working 40 hours becomes 0.027. The offer probabilities for the remaining levels of working hours remain the same. In the second simulation the entire hours offer distribution was changed: we take an hours offer distribution that is equal to the empirical distribution of desired working hours (see table 8). In the same way as in table 9, we next simulate the distribution of observed working hours. Moreover, we compute the average unemployment hazard. Note that the impact of the change in the hours offer distribution is not a priori clear and will depend on the size of the hours offer probabilities and the preferences. For instance, if the length of the working week decreases from 40 hours a week to 36 hours a week, there may be a positive impact on the hazard: if the average individual (who has optimal working hours below 40) gets a job offer, this job more likely has an hours level with a favorable number of working hours. Jobs with hours close to the optimal working hours have lower reservation wage rates (see fig. 2). Thus we may expect a positive influence on the probability of leaving unemployment. On the other hand, if jobs with hours levels close to the optimal level of working hours get a higher offer probability, the value of search rises and the unemployed will be more inclined to reject offers with a nonoptimal hours level, which may decrease the unemployment hazard.

We find that decreasing the length of the standard working week leads to a percentage increase of the hazard of 2.2%. Making the distribution of offered hours equal to the empirical distribution of desired hours increases the hazard by 1.3%. The fact that the increase in the hazard is lower in the latter case is due to the fact that if the distribution of offered hours is equal to the distribution of desired hours the value of search rises more for individuals who prefer to work less than the standard working week. Table 11 shows the impact of changing the distribution of offered hours on the distribution of accepted hours. In the first column we repeat the simulated hours distribution from table 9. The next column shows the distribution of observed hours if the standard working week has been reduced from 40 hours a week to 36 hours a week. For this purpose we exchanged the estimated hours offer probabilities of 36 and

Table 11
Policy Simulation: Impact of Changing the Length of the (Standard)
Working Week on Simulated Accepted Hours Distribution

Hours Category $h =$	Hours Offer Distribution Used in Simulation		
	Estimated Hours Offers Distribution	Hours Offer Distribution 36 Hours a Week	Hours Offer Distribution Equal to Desired Hours Distribution
Distribution of Accepted Hours (in Percentages)			
4	.1	.1	.1
8	.3	.3	.2
12	.6	.5	.3
16	1.0	.8	.5
20	2.6	1.9	2.2
24	1.4	1.0	1.6
28	1.2	.9	.9
32	3.0	2.6	10.2
36	3.2	65.2	11.7
40	65.9	3.2	57.0
44	6.7	7.3	5.7
48	2.0	2.2	1.7
52	5.7	6.6	4.8
56	1.7	2.0	1.0
60	2.8	3.3	1.5
64	.4	.5	.3
68	.3	.4	.0
72	.9	1.0	.3
76	.1	.1	.0
80	.0	.1	.1
84	.0	.0	.0

NOTE.—Hours category with hours = h : $l-2 \leq h < l+2$; $l = 4$: lower bound 0, $l = 84$: upper bound ∞ .

40 hours a week, whereas the hours offer probabilities for the remaining working hours remained the same. Due to the nature of the simulation, we see the largest effect at 36 and 40 hours: the change of the peak in the hours offer distribution from 40 to 36 hours also affects the peak of the distribution of observed hours. However, we observe additional changes in the distribution of observed working hours. In particular, we see a decrease in the hours frequencies of part-time hours in the range of 16 to 32, even though the hours offer probabilities for these levels of working hours remain the same. For instance, the probability of observing 20 hours a week is reduced by 0.7 percentage points. Apparently individuals who prefer to work part-time are more willing to accept a job with a standard length of 36 than with a standard length of 40 hours a week. Consequently, the decrease in the length of the working week induced a decrease in part-time work. We see an opposite movement at hours levels above 40 hours a week. For individuals who prefer to work more than 40 hours a week, a job with working hours of 36 is less acceptable, and therefore there is an increase in acceptances of jobs with

more than 40 hours. Since the distribution of optimal hours (see table 8) has a concentration below 40 hours a week the overall effect is that the hazard out of unemployment increases.

The final column of table 11 shows the offer distribution if the offer distribution is equal to the distribution of desired working hours (see first column of table 8). We find the shape of the offer distribution back in the simulated distribution of accepted hours, but there are also differences. The peak at 40 hours a week is more than 2 percentage points higher. Apparently, for individuals who prefer to work just below 40 hours a standard work week is also acceptable. For the individuals who want to work outside the range of 32–40 hours a week the difference between the offer distribution and the distribution of accepted hours is small. Overall, we find a smaller increase in the hazard than if the standard working week is changed.

V. Conclusions

The literature of empirical job search models usually ignores other characteristics than the wage in the job acceptance decision. In this article I present a structural job search model in which jobs are characterized by a combination that exists of a wage rate and a number of working hours that arrive randomly from a joint offer distribution. The model has a reservation utility property. The acceptance decision can also be characterized by the reservation wage rate, which is a function of working hours. The model provides a link between static labor supply models with job offer restrictions and job search models.

The distribution of observed working hours is the result of an interaction of hours restrictions originating from the demand side of the labor market and individual preferences. We use subjective data on desired working hours to disentangle the hours offer distribution and the distribution of optimal working hours, and specific care is taken to model the relationship between the observed desired hours and optimal hours. The results of three model variants are presented: a base variant that does not exploit information on desired working hours at all, the “optimal desired hours model” that follows the common practice in the literature by assuming that desired hours are equal to optimal hours, up to a measurement error, and the “hours satisfaction model” that explicitly exploits the routing and the phrasing of the questionnaire by which information on desired hours is obtained and allows for a difference between “optimality” and “satisfaction.”

The eventual empirical model is a structural job search model for unemployment duration, accepted wages and hours, and desired hours, which is estimated with data from the Dutch Socio-Economic Panel. The model allows for unobserved heterogeneity in preferences and in the job

offer arrival rate. We estimate the parameters of the utility function, and thereby optimal working hours, job offer arrival rates, layoff rates, and the wage and hours offer distribution.

All of the three before-mentioned specifications fit the data on unemployment duration about equally well according to residual analysis. The residual analysis also reveals neglected negative duration dependence for the models. Especially the longer spells of unemployment cannot be explained by a stationary model, even if unobserved heterogeneity is accounted for. The distribution of observed working hours is fitted best by the base model, but this is due to the overparameterization of the base model. The main difference in the three specifications is the way in which they succeed in disentangling hours offer probabilities from acceptance probabilities. There is a clear distinction between the base model, on the one hand, and the two specifications that allow for the use of desired hours, on the other hand.

The base specification suggests that preferences are relatively flat: this is shown by the shape of the distribution of optimal hours generated by the model as well as by the shape of the reservation wage rate as a function of hours, which does not have a pronounced minimum. In other words, it is hard to identify preferences for working hours without the use of information on desired working hours. The base model predicts that the optimal hours are on average higher than observed working hours. This is in contrast with the responses of the survey respondents, which indicate that the majority of the workers that are not satisfied with working hours report being overemployed. The optimal desired hours variant shows a concentration of optimal hours in the range of 32–44 hours a week and a pronounced minimum of the reservation wage rate at 36 hours a week. The hours satisfaction model also generates a clear relationship between reservation wage rates and working hours, even though the link between desired hours and optimal hours is much weaker in the hours satisfaction model. The hours satisfaction model differs from the optimal desired hours model as the hours satisfaction model shows smaller reservation wage rates at low working hours. This shows a larger willingness to accept jobs with part-time working hours. This is also reflected in the elasticities of the expected unemployment duration with respect to the benefit level that we computed. The hours satisfaction model shows a somewhat larger sensitivity of the hazard with respect to changes in the benefit level. Since individuals are more willing to accept jobs with working hours outside the full-time range their unemployment duration is decreasing more in response to a decrease in benefits.

If we assume that (observed) desired hours are equal to observed hours for individuals who are satisfied with their working hours we find that for the hours satisfaction model the fit of the desired hours is much better than for the optimal desired hours model. The hours satisfaction model

explains a large part of the peak that we observe in desired hours. The hours satisfaction also gives a better prediction of the overemployment that is observed in the survey responses.

The rich nature of the model is exploited to address various policy issues. Does a decrease in unemployment benefits lead to the acceptance of jobs with unfavorable hours? A model simulation in which benefits were decreased by 5% showed that the impact on accepted working hours was not large. There was a small shift in the distribution of accepted working hours toward part-time jobs. Apparently, a lower income during unemployment makes jobs with part-time hours, which have a lower income per period, more acceptable. Does a decrease in the length of the standard working week affect unemployment duration, and does it affect the distribution of accepted hours? A decrease of the standard working week was simulated by shifting the peak in the hours offer distribution down by one category, from 38–41 hours to 34–37 hours a week. Recall that a large percentage of the sample respondents report the desire to work less. The results of the simulation show that the average hazard out of unemployment is increased. There is a decrease in the percentage of workers with part-time (less than 34 hours) jobs. Apparently a job with 36 hours is more acceptable to a lot of workers with preferences for part-time jobs, and consequently they accept a new full-time job (with lower working hours) more easily. There is also a small increase in the percentage of accepted jobs with more than 40 hours a week, as for workers who want to work many hours a 36-hour working week is not acceptable anymore, whereas a 40-hour working week was. As there are more individuals who want to work less than 40 hours a week, on balance the hazard rate increases. Finally, we simulated the impact on the hazard and the accepted hours if the hours offer distribution is equal to the empirical distribution of desired hours. We also find an increase in the hazard but not as large as the increase obtained by a decrease in the length of the standard working week. If someone unemployed has a higher probability of getting a job offer with working hours equal to his desired level the value search increases, which may lead to an increase in rejections of jobs with other levels than the desired level.

Appendix A

The Reservation Utility Equation

Let $V(q)$ denote the value of search for an individual with unobserved characteristics q . While unemployed, the individual receives weekly benefits b . The amount of nonlabor income is μ . The subjective rate of dis-

count is ρ . The flow of utility, derived from benefit level b and nonlabor income μ in a short interval of length Δt is

$$u(b + \mu, 0; \epsilon) \frac{\Delta t}{1 + \rho \Delta t}. \quad (\text{A1})$$

The probability of receiving a job offer in a short interval with length Δt is $\lambda(v)\Delta t$. A job offer consists of two characteristics, wages and hours, which arrive randomly from a joint wage-hours offer distribution. The value of a job with hourly wage rate w and weekly working hours h is denoted by $W(w, h; q)$ for an individual with unobserved characteristics q . The value of a job will be compared with the value of search $V(q)$ in order to make the job acceptance decision. The value $V(q)$ for someone unemployed is

$$\begin{aligned} V(q) \approx & u(b + \mu, 0; \epsilon) \frac{\Delta t}{1 + \rho \Delta t} + \frac{1 - \lambda \Delta t}{1 + \rho \Delta t} V(q) \\ & + \frac{\lambda \Delta t}{1 + \rho \Delta t} E_{(w,h)} \max[V(q), W(w, h; q)]. \end{aligned} \quad (\text{A2})$$

After rearranging terms and taking the limit $\Delta t \rightarrow 0$ we get

$$\rho V(q) = u(b + \mu, 0; \epsilon) + \lambda E_{(w,h)} \max[0, W(w, h; q) - V(q)]. \quad (\text{A3})$$

The value of a job with wage rate w and hours h may be evaluated as

$$\begin{aligned} W(w, h; q) \approx & u(w h + \mu, h; \epsilon) \frac{\Delta t}{1 + \rho \Delta t} + \frac{\delta \Delta t}{1 + \rho \Delta t} V(q) + \frac{1 - \delta \Delta t}{1 + \rho \Delta t} W(w, h; q), \end{aligned} \quad (\text{A4})$$

in which δ is the layoff rate. Rearranging terms and letting $\Delta \rightarrow 0$ results in

$$(\rho + \delta)W(w, h; q) = u(w h + \mu, h; \epsilon) + \delta V(q). \quad (\text{A5})$$

Inserting (A5) in (A3) yields

$$\begin{aligned} \rho V(q) = & u(b + \mu, 0; \epsilon) \\ & + \frac{\lambda(v)}{\rho + \delta} E_{(w,h)} \max[0, u(w h + \mu, h; \epsilon) - \rho V(q)]. \end{aligned} \quad (\text{A6})$$

A job offer (w, h) is accepted if the utility value $u(w h + \mu, h; \epsilon)$ exceeds $\rho V(q)$. In other words, there exists a unique reservation utility level $\bar{u}(q)$ with

$$\bar{u}(q) = \rho V(q). \quad (\text{A7})$$

All job offers that generate a utility level above $\bar{u}(q)$ are acceptable; those

below will be rejected. If we assume that the marginal utility of consumption is positive $u(wb + \mu, h; \epsilon)$ is increasing in w , and therefore, for a given level of h , a wage rate $\xi(h, \bar{u}(q); \epsilon)$ can be found such that

$$u(\xi(h, \bar{u}(q); \epsilon)h + \mu, h; \epsilon) = \bar{u}(q). \quad (\text{A8})$$

This $\xi(h, \bar{u}(q); \epsilon)$ is the hours dependent reservation wage rate. Using the assumption that the wage rate and the number of working hours arrive simultaneously from the joint wage-hours offer distribution in Section II.B, equation (A6) can be rewritten such that the reservation utility equation (3) is obtained.

Appendix B

Additional Details of Likelihood Contributions

In Section II.B it was assumed that accepted wages are observed with a lognormally distributed measurement error. Formally, the relation between the observed wage w_{obs} and the offered wage w reads

$$\begin{aligned} \ln w_{\text{obs}} &= \ln w + m, \text{ conditional on } w \\ &> \xi_l(q), \quad h = h_l, \quad m \sim N(0, \sigma_m^2), \end{aligned} \quad (\text{B1})$$

in which m represents measurement error.

Equation (B1) implies that the density of the observed wage, conditional on the offered wage, reads

$$f_{\text{obs},c}(w_{\text{obs}}|w) = \frac{1}{\sqrt{2\pi}\sigma_m w_{\text{obs}}} \exp \left\{ -\frac{1}{2\sigma_m^2} [\ln w_{\text{obs}} - \ln w]^2 \right\}. \quad (\text{B2})$$

The density of an acceptable wage offer conditional on hours being equal to h_l can be derived from the job offer density:

$$\frac{f(w)}{\bar{F}_l(\xi_l(q))}, \quad l = 1, \dots, L, \quad w > \xi_l(q). \quad (\text{B3})$$

The likelihood contribution of an individual with observed unemployment duration t_u and job characteristics w_{obs} and h_l , conditional on unobserved heterogeneity, can be formed by multiplying the job offer arrival rate $\lambda(q)$, the probability p_l that hours h_l are offered, the job acceptance probability $\bar{F}_l(\xi_l(q))$, the survival probability $\exp\{-\theta(q)t_u\}$, the densities for the wages (B2) and (B3), and the density of desired hours $r(h|w, q)$. This yields the following expression:

$$\lambda(q)p_l \bar{F}_l(\xi_l(q)) \exp\{-\theta(q)t_u\} f_{\text{obs},c}(w_{\text{obs}}|w) \frac{f(w)}{\bar{F}_l(\xi_l(q))} r(h|w, q). \quad (\text{B4})$$

The accepted wage w is treated as latent and is integrated out of (B4). In

a final step, unobserved heterogeneity is integrated out. This results in the likelihood contribution (11).

The likelihood contribution (11) is for a flow sample observation. If data are collected according to a stock sampling scheme, longer unemployment durations are oversampled. In the literature, various methods to correct the likelihood for stock sampling have appeared, and some of them can be quite complicated (e.g., Nickell 1979). Ridder (1984) provides an extensive analysis of stock sample densities, and he shows that if information on backward recurrence times is available, we can condition on backward recurrence times (see also Lancaster 1979). Let r denote the elapsed unemployment duration at the time of the sampling. Then the likelihood contribution for a stock sample observation, obtained by conditioning on backward recurrence times, is

$$f(t_u, w_{\text{obs}}, h_l, \tilde{b} | t_u > r) = \frac{f(t_u, w_{\text{obs}}, h_l, \tilde{b})}{P(t_u > r)}, \quad (\text{B5})$$

with³⁷

$$P(t_u > r) = \int \exp \{-\theta(q)r\} g(q, \Sigma) dq. \quad (\text{B6})$$

Recall that we apply simulated maximum likelihood. Thus, the numerator in (B5) is replaced by (14). Similarly, we use an unbiased simulator for (B6), the denominator of (B5).

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³⁷ Note that this expression can be derived from (11).

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